

Magnetism and Electromagnetic Induction

CHAPTER 7

Electromagnetic Waves

7.1–7.10

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7

Electromagnetic Waves

INTRODUCTION

In 1865, Maxwell discovered the displacement current. This brought together the phenomenon of electricity and magnetism into a coherent and unified theory. After this discovery, Maxwell predicted the transmission of energy by electromagnetic waves, which could travel with the speed of light. This led him to conclude that the light itself is an electromagnetic wave.

According to him, an accelerated charge produces a sinusoidal time-varying magnetic field, which in turn produces a sinusoidal time-varying electric field. The two fields so produced are mutually perpendicular time-varying electric and magnetic fields. They constitute electromagnetic waves which can propagate through empty space.

The velocity of electromagnetic waves in free space is given by

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

where $\mu_0 (= 1.257 \times 10^{-6} \text{ T m A}^{-1})$ and $\epsilon_0 (= 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2})$ are, respectively, the absolute permeability and the absolute permittivity of free space. The velocity of electromagnetic waves in free space (vacuum) is equal to the velocity of light in vacuum, i.e., $3 \times 10^8 \text{ m s}^{-1}$.

ILLUSTRATION 7.1

What physical quantity is the same for X-rays of wavelength 10^{-10} m , red light of wavelength $6800 \text{ \AA}$, and radiowaves of wavelength 500 m ?

Sol. Speed is the physical quantity which is the same in vacuum for given X-ray, red light and radiowave.

Value of speed of electromagnetic waves in vacuum is

$$c = 3 \times 10^8 \text{ m s}^{-1}.$$

ILLUSTRATION 7.2

A plane electromagnetic wave travels in vacuum along z -direction. What can you say about the directions of its electric and magnetic field vectors? If the frequency of the wave is 30 MHz , what is its wavelength?

Sol. The electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$ are in the xy plane.

Using $c = \lambda \nu$, we get $\nu = \frac{c}{\lambda}$ and $\lambda = \frac{3 \times 10^8}{30 \times 60} = 10 \text{ m}$

MAXWELL'S DISPLACEMENT CURRENT

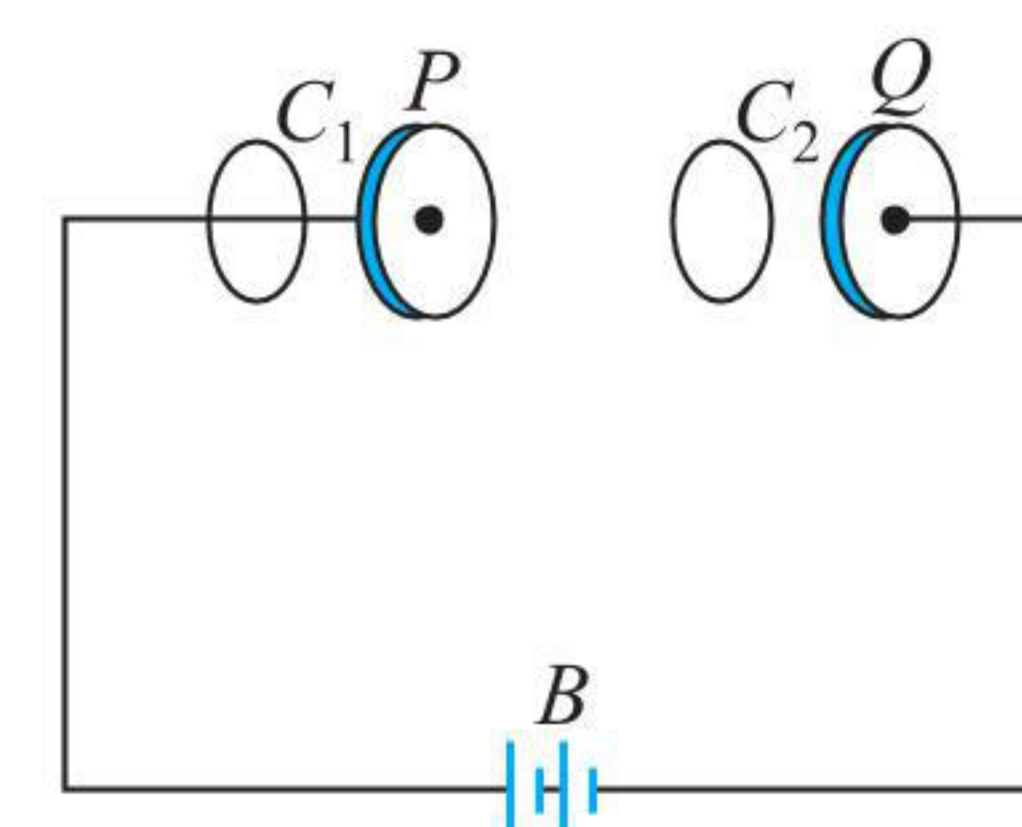
According to Ampere's circuital law, the magnetic field B is related to steady current I as

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I \quad \dots(i)$$

where I is the current travelling the surface bounded by closed path C .

In 1864, Maxwell showed that relation (i) is logically inconsistent. He accounted for this inconsistency as follows:

Consider a parallel plate capacitor having plates P and Q being charged with battery B .



During charging, a current I flows through the connecting wires which changes with time. This current will produce magnetic field around the wires which can be detected using a magnetic compass needle.

Consider two loops c_1 and c_2 parallel to the plates P and Q of the capacitor. c_1 is enclosing only the connecting wire attached to the plate P of the capacitor and c_2 lies in the region between the two plates of capacitor. For the loop c_1 , a current I is flowing through it, hence Ampere's circuital law for loop c_1 gives

$$\oint_{c_1} \vec{B} \cdot d\vec{l} = \mu_0 I \quad \dots(ii)$$

Since the loop c_2 lies in the region between the plates of the capacitor, no current flows in this region. Hence Ampere's circuital law for loop c_2 gives

$$\oint_{c_2} \vec{B} \cdot d\vec{l} = 0 \quad \dots(iii)$$

The relations (ii) and (iii) continue to be true even if two loops c_1 and c_2 are infinitesimally close to the plate P of the capacitor. On the other hand, as the loops c_1 and c_2 are infinitesimally close, it is expected that

$$\oint_{c_1} \vec{B} \cdot d\vec{l} = \oint_{c_2} \vec{B} \cdot d\vec{l} \quad \dots(iv)$$

Thus, relation (iv) is in contradiction with relations (ii) and (iii). This led Maxwell to point out that Ampere's circuital law as given by (i) is logically inconsistent.

Idea of Displacement Current: Maxwell predicted that not only a current flowing in a conductor produces magnetic field but also a time-varying electric field (i.e., changing electric field) in a vacuum/free space (or in a dielectric) produces a magnetic field.

It means a changing electric field gives rise to a current which flows through a region so long as the electric field is changing there. Maxwell also predicted that this current produces the same magnetic field as a conduction current can produce. This current is known as 'displacement current'.

Thus, displacement current is that current which comes into play in the region in which the electric field and hence the electric flux is changing with time.

Maxwell defined this displacement current in space where electric field is changing with time as

$$I_D = \epsilon_0 \frac{d\phi_E}{dt} \quad \dots(v)$$

where ϕ_E is the electric flux.

Maxwell also found that conduction current (I) and displacement current (I_D) together have the property of continuity, although, individually, they may not be continuous.

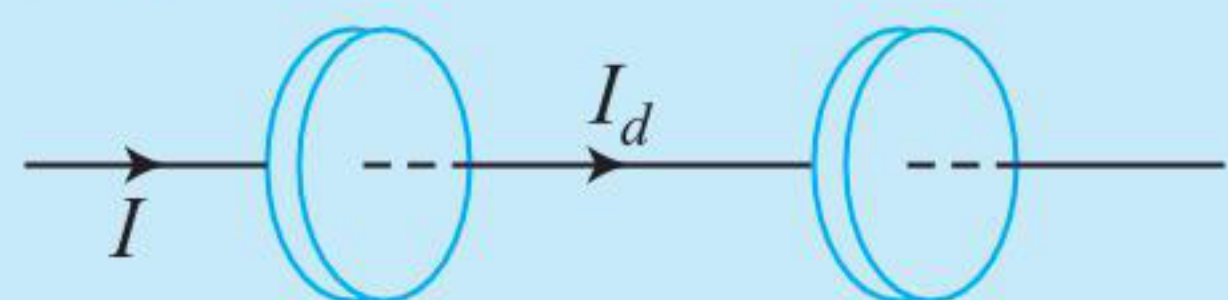
This idea led Maxwell to modify Ampere's circuital law in order to make the same logically consistent. He states Ampere circuital law to the form,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0(I + I_D) = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

It is now called as Ampere Maxwell law.

ILLUSTRATION 7.3

The following figure shows capacitor made of two circular plates. The capacitor is being charged by an external source which supply constant current equal to 0.15 A. Obtain displacement charged by constant current across plates. Given $\frac{dV}{dt} = 1.87 \times 10^9 \text{ V s}^{-1}$.



Sol. Displacement current $I_d = \frac{\epsilon_0 d\phi_C}{dt}$

$$\text{But } \phi_E = EA = \frac{d}{\epsilon_0 A} A = \frac{q}{\epsilon_0}$$

$$\therefore I_d = \epsilon_0 \frac{d}{dt} \frac{q}{\epsilon_0} = \frac{\epsilon_0}{\epsilon_0} \frac{dq}{dt} = \frac{dq}{dt} = I$$

$$\text{Here } I = 0.15 \text{ A} \quad \therefore I_d = 0.15 \text{ A}$$

MAXWELL'S EQUATIONS

The four Maxwell's equations and Lorentz force law together constitute the foundations of classical electromagnetism. The Maxwell's equations are:

$$1^{\text{st}} \text{ equation: } \oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

$$2^{\text{nd}} \text{ equation: } \oint \vec{B} \cdot d\vec{S} = 0$$

$$3^{\text{rd}} \text{ equation: } \oint \vec{B} \cdot d\vec{l} = \frac{-d}{dt} \iint \vec{B} \cdot d\vec{S}$$

$$4^{\text{th}} \text{ equation: } \oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d}{dt} \iint \vec{B} \cdot d\vec{S} + \mu_0 \cdot I$$

The Lorentz equation is

$$\vec{F} = q (\vec{E} \times \vec{v} \times \vec{B})$$

$$E_z = E_0 \sin(\omega t - ky)$$

$$B_x = B_0 \sin(\omega t - ky), \text{ where } \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Since $\omega = 2\pi f$, where f is the frequency and $k = 2\pi/\lambda$, where λ is the wavelength.

$$\text{Therefore, } \frac{\omega}{k} = \frac{2\pi f}{2\pi/\lambda} = f\lambda$$

But $f\lambda$ gives the velocity of the wave. Hence $f\lambda = c = \omega/k$. Hence, the velocity of the waves is given by

$$c = \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

ILLUSTRATION 7.4

Suppose that the electric field amplitude of an electromagnetic wave is $E_0 = 120 \text{ N/C}$ and that its frequency is $\nu = 50.0 \text{ MHz}$.

(a) Determine B_0 , ω , k , and λ . (b) Find expressions for $\vec{E}$ and $\vec{B}$.

Sol.

$$(a) (i) \frac{E_{\text{max}}}{B_{\text{max}}} = c, \text{ we get}$$

$$B_{\text{max}} = \frac{E}{c} = \frac{120}{3 \times 10^8} = 40 \times 10^{-7} \text{ T}$$

$$\text{i.e., } B_{\text{max}} = 400 \text{ nT}$$

$$(ii) \omega = 2\pi\nu = 2 \times \pi \times 50 \times 10^6 = 3.14 \times 10^8 \text{ rad s}^{-1}$$

$$(iii) k = \frac{2\pi}{\lambda} = \frac{2\pi}{6} = \frac{2 \times 3.14}{6} = 1.05 \text{ rad s}^{-1}$$

$$(iv) c = \lambda\nu \text{ or } \lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{50 \times 10^6} = 6 \text{ m}$$

$$(b) \vec{E} = E_{\text{max}} \sin(kx - \omega t) \hat{j} = 120 \sin(1.05x - 3.14 \times 10^8 \times t) \hat{j}$$

$$\vec{B} = B_{\text{max}} \sin(kx - \omega t) \hat{k} = 400 \sin(1.05x - 3.14 \times 10^8 \times t) \hat{k}$$

ILLUSTRATION 7.5

Suppose that the electric field part of a electromagnetic wave in vacuum is

$$\vec{E} = (3.1 \text{ N/C}) \cos(1.8 \text{ rad/m}y + (5.4 \times 10^6 \text{ rad/s})t)$$

(a) What is the direction of propagation?

(b) What is the wavelength λ ?

(c) What is the frequency ν ?

(d) What is the amplitude of the magnetic field part of the wave?

(e) Write an expression for the magnetic field part of the wave.

Sol. Here $\vec{E} = 3.1 \cos \{1.8y + (5.4 \times 10^6)t\} \hat{i}$

Comparing it with standard equation

$$\vec{E} = E_0 \cos(kx + \omega t) \hat{i}$$

(a) $-\hat{j}$ direction

(b) Using $k = \frac{2\pi}{\lambda}$, we get $\lambda = \frac{2\pi}{k} = \frac{2\pi}{1.8} = 3.5 \text{ m}$

(c) Frequency, $\nu = \frac{\omega}{2\pi} = \frac{5.4 \times 10^6}{2\pi} = 0.86 \text{ MHz}$

(d) $c = \frac{E}{B}$ or $B = \frac{E}{c} = \frac{3.1}{3 \times 10^8} = 10 \text{ nT}$

(e) $\vec{B} = 10 \cos(1.8y + 5.4 \times 10^6 t) \hat{k}$

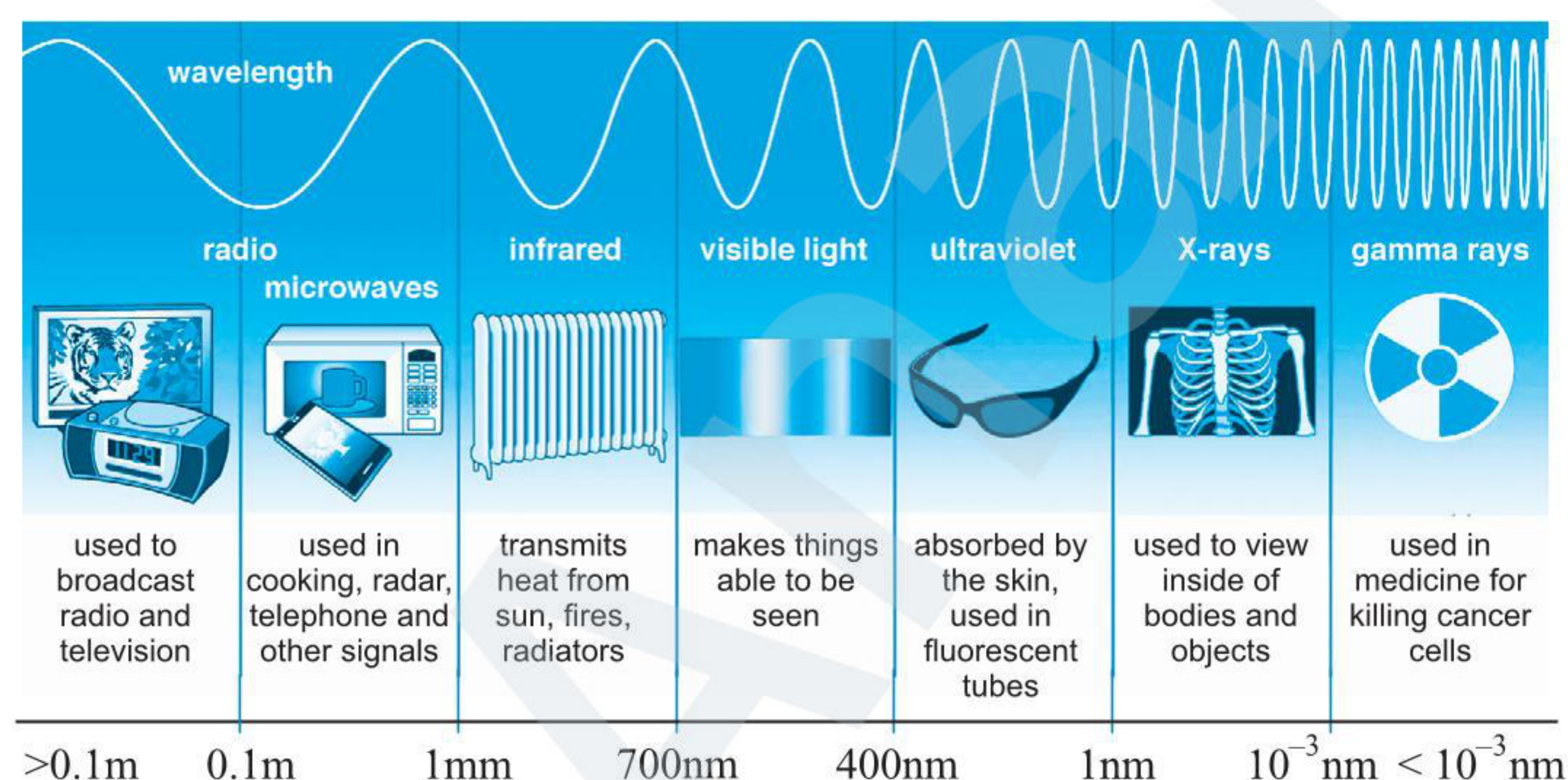
ELECTROMAGNETIC WAVE SPECTRUM

All the known radiations form a big family of electromagnetic wave which stretches over a large range of wavelengths. The orderly distribution of the electromagnetic waves in accordance with their wavelength or frequency into **distinct** groups having widely differing properties is **called** electromagnetic spectrum.

The wavelengths in the range of nearly 380 to 700 nm, produce sensation of vision and are usually called as light. The electromagnetic radiations having higher frequencies than the visible light are ultraviolet rays, X-rays and gamma rays. Those having frequencies less than that of visible light are infrared, microwaves and radio waves.

The various regions of the e.m. spectrum do not have sharply defined boundaries and they overlap. The classification is based roughly on how the waves are produced and/or detected.

As shown in figure, the main parts of the e.m. spectrum are γ -rays, X-rays, ultraviolet rays, visible light, infrared rays, microwaves and radio waves in the order of increasing wavelength from $10^{-2} \text{ \AA}$ or 10^{-12} m to 10^6 m .



1. Radio waves: These are the e.m. waves of longest wavelength and minimum frequency.

Wavelength range	600 m to 0.1 m
Frequency range	500 kHz to 1000 MHz
Source	Accelerated motion of charges in conducting wires or oscillating circuits.
Properties	Reflection, diffraction

Use of radio waves:

- In radio and television communication systems.
- In radio astronomy.

Some important wireless communication bands

Frequency band	Service
540–1600 kHz	Medium wave AM band
3–30 MHz	Shortwave AM band
88–108 MHz	FM broadcast
54–890 MHz	TV waves
840–935 MHz	Mobile phones

2. Microwaves: These are the e.m. waves having wavelengths next smaller to radio waves.

Wavelength range	0.1 m to 10^{-3} m
Frequency range	10^9 Hz to 10^{12} Hz
Source	Oscillating currents in special vacuum tubes like klystrons, magnetrons and Gunn diodes.
Properties	Reflection, refraction, diffraction and polarization. Due to their shorter wavelengths, they can travel as a beam in a signal.

Uses of microwaves:

Microwaves have shorter wavelengths than radio waves. This causes less diffraction when they encounter buildings, trees, hills etc. This makes them suitable for radars. They require smaller antenna compared to the radio waves. Microwaves can be used:

- In radar systems for aircraft navigation
- In long-distance communication systems via geostationary satellites
- In microwave ovens

3. Infrared waves: These waves have wavelengths greater than the red end of the visible spectrum. Infrared waves produce heating effect, so they are also known as heat waves or thermal radiation.

Wavelength range	$5 \times 10^{-3} \text{ m}$ to 10^{-6} m
Frequency range	10^{11} Hz to $5 \times 10^{14} \text{ Hz}$
Source	Hot bodies and molecules
Properties	Heating effect, reflection, refraction, diffraction and propagation through fog

Uses of infrared waves:

- In the remote control of a TV or VCR, the keypad of which contains a small infrared transmitter.
- In green houses to keep the plants warm.
- In haze photography because infrared waves are less scattered than visible light by atmospheric particles.
- Infrared lamps in the treatment of muscular complaints.
- In reading the secret writings on the ancient walls.
- In knowing the molecular structure.

4. **Visible light:** It is a very small part of the e.m. spectrum towards which the human retina is sensitive. The visible light emitted or reflected from bodies around us gives information about the world.

Wavelength range	$8 \times 10^7 \text{ m}$ to $4 \times 10^{-7} \text{ m}$
Frequency range	$4 \times 10^{14} \text{ Hz}$ to $7 \times 10^{14} \text{ Hz}$
Source	Radiated by excited atoms in ionized gas and incandescent bodies
Properties	Reflection, refraction, interference, diffraction, polarization, photoelectric effect, photographic action, sensation of sight

The approximate wavelength ranges for lights of different colours are as follows:

Visible Spectrum

Colour	Wavelength range	Colour	Wavelength range
Violet, indigo	4000–4500 Å	Yellow	5700–5900 Å
Blue	4500–5000 Å	Orange	5900–6200 Å
Green	5000–5700 Å	Red	6200–7500 Å

5. **Ultraviolet light:** These are electromagnetic waves having wavelengths shorter than the visible violet light and can be detected just beyond the violet end of the solar spectrum.

The Sun sends a lot of ultraviolet radiation to us. But fortunately, most of it is absorbed by the ozone layer in the atmosphere. A large exposure of UV rays is harmful for us. UV radiation is effectively absorbed by simple glass. Hence, one cannot get sunburn through a glass window. Welders wear special glass goggles and glass face mask to protect them from the UV rays produced in the welding arc.

Wavelength range	$3.5 \times 10^{-7} \text{ m}$ to $1.5 \times 10^{-7} \text{ m}$
Frequency range	10^{16} Hz to 10^{17} Hz
Source	High voltage gas discharge tubes, arcs of iron and mercury, the sun
Properties	Effect on photographic plate, fluorescence, ionisation, highly energetic tanning of the human skin

Uses of ultraviolet light:

- In food preservation.
- In the study of invisible writings, forged documents and finger prints.
- In the study of molecular structure.

6. **X-rays:** These e.m. waves have wavelengths just shorter than ultraviolet light. X-rays have high penetration power. They are used in detecting bone fractures, so they have many useful medical and industrial applications. Excessive exposure to X-ray is dangerous.

Wavelength range	100 Å to 0.1 Å
Frequency range	10^{18} Hz to 10^{20} Hz
Source	Sudden deceleration of fast moving electrons by a metal target
Properties	Effect on photographic plate, ionization of gases, photoelectric effect, fluorescence, more energetic than UV rays

Uses of X-rays:

- In the study of crystals structure because X-rays can be reflected and diffracted by crystals.
- In engineering for detecting faults, cracks, flaws and holes in the finished metal products.
- In detective departments to detect explosive, diamond, gold, etc. in the possession smugglers.
- In radiotherapy to cure untraceable skin diseases and malignant growths.

7. **Gamma rays:** These are e.m. radiations of highest frequency range and lowest wavelength range. These are most penetrating e.m. waves.

Wavelength range	10^{-14} m to 10^{-10} m
Frequency range	10^{18} Hz to 10^{22} Hz
Source	Radioactive nuclei and nuclear reactions. Co-60 is a pure γ -ray source
Properties	Effect on photographic plate, fluorescence, ionization, diffraction, high penetrating power

Uses of γ -rays:

- In radiotherapy for the treatment of malignant tumours.
- In the manufacture of polyethylene from ethylene.
- To initiate some nuclear reactions.
- To preserve food stuffs for a long time because soft γ -rays can kill microorganisms.
- To study the structure of atomic nuclei.

ENERGY DENSITY

The electric and magnetic fields in a plane electromagnetic wave are given by

$$E = E_0 \sin \omega(t - x/c)$$

and $B = B_0 \sin \omega(t - x/c)$

In any small volume dV , the energy of the electric field is

$$U_E = \frac{1}{2} \epsilon_0 E^2 dV$$

and the energy of the magnetic field is $U_B = \frac{1}{2\mu_0} B^2 dV$

Thus, the total energy is $U = \frac{1}{2} \epsilon_0 E^2 dV + \frac{1}{2\mu_0} B^2 dV$

The energy density is $u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2$

$$u = \frac{1}{2} \epsilon_0 E_0^2 \sin^2 \omega(t - x/c) + \frac{1}{2\mu_0} B_0^2 \sin^2 \omega(t - x/c)$$

If we take the average over a long time, the $\sin^2$ terms have an average value of $1/2$

$$\text{Thus, } u_{av} = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4\mu_0} B_0^2$$

Now, $E_0 = cB_0$ and $\mu_0 \epsilon_0 = \frac{1}{c^2}$ so that, $\mu_0 = \frac{1}{\epsilon_0 c^2}$ and

$$B_0 = \frac{E_0}{c} \quad \frac{1}{4\mu_0} B_0^2 = \frac{\epsilon_0 c^2}{4} \left(\frac{E_0}{c} \right)^2 = \frac{1}{4} \epsilon_0 E_0^2$$

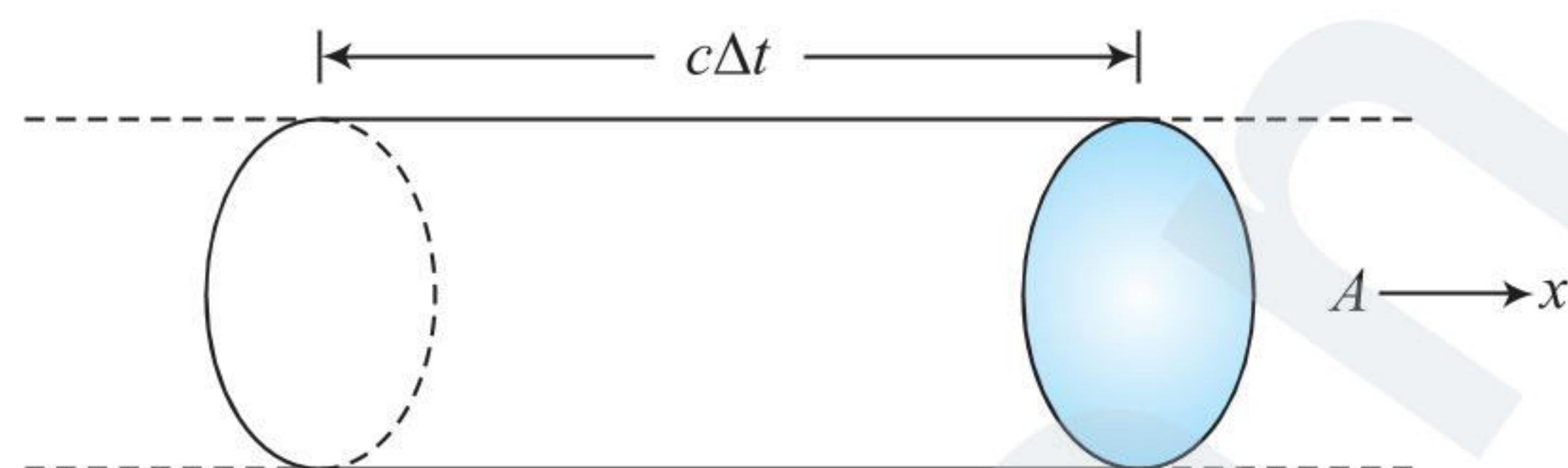
Thus, the electric energy density is equal to the magnetic density in average,

$$\text{or, } u_{av} = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4} \epsilon_0 E_0^2 = \frac{1}{2} \epsilon_0 E_0^2$$

$$\text{Also } u_{av} = \frac{1}{4\mu_0} B_0^2 + \frac{1}{4\mu_0} B_0^2 = \frac{1}{2\mu_0} B_0^2$$

INTENSITY

The energy crossing per unit area per unit time perpendicular to the direction of propagation is called the intensity of a wave.



Consider a cylindrical volume with area of cross section A and length $c \Delta t$ along the x -axis. The energy contained in this cylinder crosses the area A in time Δt as the wave propagates at speed c . The energy contained is

$$U = u_{av} (c \Delta t) A$$

The intensity is $I = \frac{U}{A \Delta t} = u_{av} c$ in terms of maximum electric field, $I = \frac{1}{2} \epsilon_0 E_0^2 c$

MOMENTUM

The electromagnetic wave also carries linear momentum with it. The linear momentum carried by the portion of wave having energy U is given by $p = U/c$.

Thus, if the wave incident on a material surface is completely absorbed, it delivers energy U and momentum $p = U/c$ to the surface. If the wave is totally reflected, the momentum delivered is $2U/c$ because the momentum of the wave changes from p to

$-p$. It follows that electromagnetic waves incident on a surface exert a force on the surface.

PROPERTIES OF ELECTROMAGNETIC WAVES

- In electromagnetic waves the electric field vector $\vec{E}$, magnetic field vector $\vec{B}$ and propagation vector $\vec{k}$ are mutually perpendicular such that $\vec{E}$, $\vec{B}$, and $\vec{k}$ form a right angle system. Hence electromagnetic waves are transverse in nature.
- Electromagnetic waves travel with speed of light. In vacuum their speed is

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

In isotropic medium, their speed is

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_r \mu_0 \epsilon_r \epsilon_0}} = \frac{1}{\sqrt{\mu_r \epsilon_r}} \cdot \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{c}{n}$$

where $n = \sqrt{\mu_r \epsilon_r}$ is the refractive index of the medium, μ_r = relative permeability of medium, and ϵ_r = relative permittivity of medium or electric dielectric constant.

- The Poynting vector $\vec{S} = \vec{E} \times \vec{H}$ represents the direction of energy flow per unit area per second along the direction of wave propagation. Its unit is Wm^{-2} .
- The medium offers hindrance to the propagation of wave. Such a hindrance is called Wave Impedance (Z) and its value in a medium is given by

$$Z = \frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_r}{\epsilon_r}} \sqrt{\frac{\mu_0}{\epsilon_0}} \\ (\because \mu = \mu_r \mu_0 \text{ and } \epsilon = \epsilon_r \epsilon_0)$$

where μ_r and ϵ_r are relative permeability and relative permittivity of the medium.

For vacuum or free space

$$Z = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.6 \, \Omega$$

Also in free space

$$\frac{E}{B} = \frac{E}{\mu_0 H} = \frac{1}{\mu_0} \sqrt{\frac{\mu_0}{\epsilon_0}} \\ \frac{E}{B} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c \\ E = Bc$$

- In free space
The electrostatic energy density is equal to magnetostatic energy density.
Average electric energy density = Average magnetic energy density

$$= \frac{1}{4} \epsilon_0 E_0^2 = \frac{1}{4} \mu_0 H_0^2 = \frac{B_0^2}{4\mu_0}$$

Total average energy density = $\frac{1}{2} \epsilon_0 E_0^2$ (in free space)

(f) In a medium

Electric energy density = $\frac{1}{2} \epsilon E^2$

Magnetic energy density = $\frac{B^2}{2\mu}$

Total energy density = $\frac{1}{2} \epsilon E^2 + \frac{B^2}{2\mu}$

(g) The electric and magnetic fields satisfy the following wave equations, which can be obtained from Maxwell's third and fourth equations.

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2}$$

(h) Electromagnetic waves carry momentum and hence can exert pressure (P) on surfaces, which is known as radiation pressure. For an electromagnetic wave with Poynting vector $\vec{S}$, incident upon a perfectly absorbing surface.

$$P = \frac{S}{c}$$

and if incident upon a perfectly reflecting surface, then

$$P = \frac{2S}{c}$$

(i) The electric and magnetic fields of a sinusoidal plane electromagnetic wave propagating in the positive x -direction can also be written as

$$E = E_0 \sin(kx - \omega t)$$

$$B = B_0 \sin(kx - \omega t)$$

where ω is the angular frequency of the wave and k is the angular wave number or propagation constant which are given by

$$\omega = 2\pi f \text{ and } k = \frac{2\pi}{\lambda}$$

(j) The intensity of a sinusoidal plane electromagnetic wave is defined as the average value of Poynting vector taken over one cycle.

ATMOSPHERE

The atmosphere of earth extends upon about 300 km. The composition of atmosphere is as given below.

TROPOSPHERE

Troposphere extends up to a height of about 12 km. This layer consists of water droplets, vapour, dust particles etc. Density of air decreases as we move from bottom to top of this layer. This layer is responsible for all the weather phenomena affecting our environment.

STRATOSPHERE

This layer extends from about 10 km to about 50 km. In the upper part of the atmosphere, we have a layer of ozone. The density of

air at the top of the stratosphere is about 10^{-3} times the density at the surface of earth.

MESOSPHERE

The mesosphere extends from about 50 km to about 80 km.

IONOSPHERE

The atmosphere above the mesosphere is called ionosphere. This layer is composed partly of electrons and positive ions. The rest of the atmosphere is composed mostly of neutral molecules.

The atmosphere is transparent to visible radiation and we can see the sun and the stars through it clearly. However, most infrared radiation is not able to pass through, as it is absorbed by the atmosphere. Low lying clouds in the atmosphere also prevent infrared radiation from passing through. The ozone layer blocks the passage of ultraviolet radiation from the sun.

ILLUSTRATION 7.6

About 5% of the power of a 100 W light bulb is converted to visible radiation. What is the average intensity of visible radiation

- (a) At a distance of 1 m from the bulb?
(b) At a distance of 10 m?

Assume that the radiation is emitted isotropically and neglect reflection.

Sol.

(a) Visible power = 5 W

$\therefore$ Average intensity of radiation at 1 m

$$= \frac{\text{Power}}{4\pi r^2} = \frac{5}{4 \times \pi \times 1} = 0.4 \text{ W m}^{-2}$$

(b) Average intensity of radiation at 10 m

$$= \frac{\text{Power}}{4\pi r^2} = \frac{5}{4 \times \pi \times 10^2} = 0.004 \text{ W m}^{-2}$$

ILLUSTRATION 7.7

Given below are some famous numbers associated with electromagnetic radiation in different contexts in physics. State the part of the em spectrum of which each belongs.

- (a) 21 cm (wavelength emitted by atomic hydrogen in interstellar space)
(b) 1057 MHz (frequency of radiation arising from two close energy levels in hydrogen, known as Lamb shift)
(c) 2.7 K (temperature associated with the isotropic radiation filling all space—thought to be a relic of the ‘big-bang’ origin of the universe)
(d) 5890 Å–58963 Å (double lines of sodium)
(e) 14.4 keV (energy of a particular transition in ^{57}Fe nucleus associated with a famous high resolution spectroscopic method of sodium)

Sol.

(a) Wavelength is of the order of 10^{-2} m, i.e., short radio wave.

(b) Frequency is of the order of 10^{-2} Hz, i.e., short radio wave.

(c) $\lambda_m T = 0.29$ or

$$\lambda_m = \frac{0.29}{T} = \frac{0.29}{2.7} = 0.09 \text{ cm} = 0.0009 \text{ m}$$

Wavelength is of the order of 10^{-4} m, i.e., microwave.

(d) Wavelengths are of the order of 10^{-7} m, i.e., visible radiations.

(e) $\text{Energy} = \frac{hc}{\lambda e}$ or $\lambda = \frac{hc}{\text{Energy} \times e}$

$$\lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{14.4 \times 10^3 \times 1.6 \times 10^{-19}}$$

$$\text{i.e., } \lambda = 0.86 \times 10^{-10}$$

$$= 8.6 \times 10^{-11} \text{ m}$$

i.e., X-rays.

Exercises

Single Correct Answer Type

- The intensity of sunlight (in W/m^2) at the solar surface will be
 (1) 5.6×10^6 (2) 5.6×10^7
 (3) 4.2×10^6 (4) 4.2×10^7
- Which of the following has zero average value in a plane electromagnetic wave?
 (1) Electric field (2) Magnetic potential
 (3) Electric energy (4) Magnetic energy
- The wave impedance of free space is
 (1) 0Ω (2) 376.6Ω
 (3) 1883Ω (4) 3776Ω
- The frequency 1057 MHz of radiation arising from two close energy levels in hydrogen belongs to
 (1) radio waves (2) infrared waves
 (3) micro waves (4) γ -rays
- The wave associated with 2.7 K belongs to
 (1) radio waves (2) micro waves
 (3) ultraviolet ways (4) infrared waves
- The percentage power of X-ray increases with the increases in its
 (1) velocity (2) intensity
 (3) frequency (4) wavelength
- The shortest wavelength of X-rays remitted from an X-ray tube depends upon
 (1) nature of the gas in the tube
 (2) voltage applied to the tube
 (3) current in the tube
 (4) nature of target of the tube
- X-rays are not used for radar purposes, because they are not
 (1) reflected by target
 (2) partly absorbed by target
 (3) electromagnetic waves
 (4) completely absorbed by target
- The energy of X-rays photon is $3.3 \times 10^{-16} \text{ J}$. Its frequency is
 (1) $2 \times 10^{19} \text{ Hz}$ (2) $5 \times 10^{18} \text{ Hz}$
 (3) $5 \times 10^{17} \text{ Hz}$ (4) $5 \times 10^{16} \text{ Hz}$
- In X-ray tube, the accelerating potential applied at the anode is V volt. The minimum wavelength of the emitted X-ray will be
 (1) eV/h (2) h/eV
 (3) eV/ch (4) hc/eV
- The area of television telecast is made twice, the height of antenna will be changed as
 (1) halved (2) doubled
 (3) quadrupled (4) kept unchanged
- If a source is transmitting electromagnetic wave of frequency $8.2 \times 10^6 \text{ Hz}$, then the wavelength of the electromagnetic waves transmitted from the source will be
 (1) 36.6 m (2) 40.5 m
 (3) 42.3 m (4) 50.9 m
- A plane electromagnetic wave
 $E_z = 100 \cos(6 \times 10^8 t + 4x) \text{ V/m}$
 propagates in a medium of dielectric constant
 (1) 1.5 (2) 2.0
 (3) 2.4 (4) 4.0
- If μ_0 be the permeability and K_0 the dielectric constant of a medium, its refractive index is given by
 (1) $\frac{1}{\sqrt{\mu_0 K_0}}$ (2) $\frac{1}{\mu_0 K_0}$
 (3) $\sqrt{\mu_0 K_0}$ (4) $\mu_0 K_0$
- If ϵ_0 and μ_0 represent the permittivity and permeability of vacuum and ϵ and μ represent the permittivity and permeability of medium, the refractive index of the medium is given by
 (1) $\sqrt{\frac{\epsilon_0 \mu_0}{\epsilon \mu}}$ (2) $\sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}}$
 (3) $\sqrt{\frac{\epsilon}{\mu_0 \epsilon_0}}$ (4) $\sqrt{\frac{\mu_0 \epsilon_0}{\epsilon}}$
- In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of $2.5 \times 10^{10} \text{ Hz}$ and amplitude 480 V/m. The amplitude of the oscillating magnetic field will be
 (1) $1.52 \times 10^{-8} \text{ Wb/m}^2$ (2) $1.52 \times 10^{-7} \text{ Wb/m}^2$
 (3) $1.6 \times 10^{-6} \text{ Wb/m}^2$ (4) $1.6 \times 10^{-7} \text{ Wb/m}^2$
- The magnetic field between the plates of radius 12 cm separated by a distance of 4 mm of a parallel plate capacitor of capacitance 100 pF along the axis of plates having conduction current of 0.15 A is
 (1) zero (2) 1.5 T
 (3) 15 T (4) 0.15 T
- A flood light is covered with a filter that transmits red light. The electric field of the emerging beam is represented by a sinusoidal plane wave
 $E_x = 36 \sin(1.20 \times 10^7 z - 3.6 \times 10^{15} t) \text{ V/m}$
 The average intensity of beam in W/m^2 will be
 (1) 6.88 (2) 3.44
 (3) 1.72 (4) 0.86
- The average energy-density of electromagnetic wave given by $E = (50 \text{ N/C}) \sin(\omega t - kx)$ will be nearly
 (1) 10^{-8} J/m^3 (2) 10^{-7} J/m^3
 (3) 10^{-6} J/m^3 (4) 10^{-5} J/m^3
- A larger parallel plate capacitor, whose plates have an area of 1 m^2 are separated from each other by 1 mm. If the plates has dielectric constant 10, then the displacement current at this instant is
 (1) 25 μA (2) 11 μA
 (3) 2.2 μA (4) 1.1 μA

21. A parallel plate capacitor with plate area A and separation between the plates d , is charged by a constant current i . Consider a plane surface of area $A/2$ parallel to the plates and drawn simultaneously between the plates. The displacement current through this area is
 (1) i (2) $i/2$
 (3) $i/4$ (4) $i/8$
22. The sun delivers 10^4 W/m^2 of electromagnetic flux to the earth's surface. The total power that is incident on a roof of dimensions $(10 \times 10) \text{ m}^2$ will be
 (1) 10^4 W (2) 10^5 W
 (3) 10^6 W (4) 10^7 W
23. The average value of electric energy density in an electromagnetic wave is (E_0 is peak value)
 (1) $\frac{1}{2} \epsilon_0 E_0^2$ (2) $\frac{E_0^2}{2\epsilon_0}$
 (3) $\epsilon_0 E_0^2$ (4) $\frac{1}{4} \epsilon_0 E_0^2$
24. The sun delivers 10^3 W/m^2 of electromagnetic flux to the earth's surface. The total power that is incident on a roof of dimensions $8 \text{ m} \times 20 \text{ m}$ will be
 (1) $2.56 \times 10^4 \text{ W}$ (2) $6.4 \times 10^5 \text{ W}$
 (3) $4.0 \times 10^5 \text{ W}$ (4) $1.6 \times 10^5 \text{ W}$
25. In Q. 24, the radiation force on the roof will be
 (1) $8.53 \times 10^{-5} \text{ N}$ (2) $2.3 \times 10^{-3} \text{ N}$
 (3) $1.33 \times 10^{-3} \text{ N}$ (4) $5.33 \times 10^{-4} \text{ N}$
26. A plane electromagnetic wave of wave intensity 6 W/m^2 strikes a small mirror of area 39 cm^2 , held perpendicular to the approaching wave. The momentum transferred in kg ms^{-1} by the wave to the mirror each second will be
 (1) 1.2×10^{-10} (2) 2.4×10^{-9}
 (3) 3.6×10^{-8} (4) 4.8×10^{-7}
27. The wave of wavelength $5900 \text{ \AA}$ emitted by any atom or molecule must have some finite total length which is known as coherence length. For sodium light, this length is 2.4 cm . The number of oscillations in this length will be
 (1) 4.068×10^8 (2) 4.068×10^4
 (3) 4.068×10^6 (4) 4.068×10^5
28. A radiowave has a maximum electric field intensity of 10^{-4} V m^{-1} on arrival at a receiving antenna. The maximum magnetic flux density of such a wave is
 (1) zero (2) $3 \times 10^4 \text{ T}$
 (3) $5.8 \times 10^{-9} \text{ T}$ (4) $3.3 \times 10^{-13} \text{ T}$
29. The transmitting antenna of a radio-station is mounted vertically. At a point 10 km due north of the transmitter the peak electric field is 10^{-3} V/m . The amplitude of the radiated magnetic field is
 (1) $3.33 \times 10^{-10} \text{ T}$ (2) $3.33 \times 10^{-12} \text{ T}$
 (3) 10^{-3} T (4) $3 \times 10^5 \text{ T}$
30. A circular ring of radius r is placed in a homogeneous magnetic field perpendicular to the plane of the ring. The field B changes with time according to the equation $B = Kt$, where K is a constant and t is the time. The electric field in the ring is
 (1) $\frac{Kr}{4}$ (2) $\frac{Kr}{3}$
 (3) $\frac{Kr}{2}$ (4) $\frac{K}{2r}$
31. A plane em wave of wave intensity of 10 W/m^2 strikes a small mirror of area 20 cm^2 , held perpendicular to the approaching wave. The radiation force on the mirror will be
 (1) $6.6 \times 10^{-11} \text{ N}$ (2) $1.33 \times 10^{-11} \text{ N}$
 (3) $1.33 \times 10^{-10} \text{ N}$ (4) $6.6 \times 10^{-10} \text{ N}$
32. In a region of free space the electric field at some instant of time is $\vec{E} = (80\hat{i} + 32\hat{j} - 64\hat{k}) \text{ V/m}$ and the magnetic field is $\vec{B} = (0.2\hat{i} + 0.08\hat{j} - 0.29\hat{k}) \mu\text{T}$. The pointing vector for these fields is
 (1) $-11.52\hat{i} + 28.8\hat{j}$ (2) $-28.8\hat{i} + 11.52\hat{j}$
 (3) $28.8\hat{i} - 11.52\hat{j}$ (4) $11.52\hat{i} - 28.8\hat{j}$
33. A plane electromagnetic wave propagating in the x -direction has wavelength of 60 mm . The electric field is in the y -direction and its maximum magnitude is 33 V/m^{-1} . The equation for the electric field as function of x and t is
 (1) $11 \sin \pi(t - x/c)$ (2) $33 \sin \pi \times 10^{11}(t - x/c)$
 (3) $33 \sin \pi(t - x/c)$ (4) $11 \sin \pi \times 10^{11}(t - x/c)$
34. In an electromagnetic wave, the electric field oscillated sinusoidally with amplitude 45 Vm^{-1} , the rms value of oscillating magnetic field will be
 (1) $1.6 \times 10^{-8} \text{ T}$ (2) $16 \times 10^{-9} \text{ T}$
 (3) $144 \times 10^{-8} \text{ T}$ (4) $11.3 \times 10^{-8} \text{ T}$
35. The magnetic field in the plane electromagnetic wave is given by
 $B_z = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \text{ tesla}$.
 The expression for electric field will be
 (1) $E_z = 30\sqrt{2} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \text{ V/m}$
 (2) $E_z = 60 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \text{ V/m}$
 (3) $E_y = 30\sqrt{2} \sin(0.5 \times 10^{11} x + 0.5 \times 10^3 t) \text{ V/m}$
 (4) $E_y = 60 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \text{ V/m}$

Answers Key

EXERCISES

Single Correct Answer Type

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (2) | 2. (1) | 3. (3) | 4. (1) | 5. (2) |
| 6. (3) | 7. (2) | 8. (1) | 9. (3) | 10. (4) |
| 11. (2) | 12. (1) | 13. (2) | 14. (3) | 15. (2) |

- | | | | | |
|---------|---------|---------|---------|---------|
| 16. (3) | 17. (1) | 18. (3) | 19. (1) | 20. (3) |
| 21. (2) | 22. (3) | 23. (4) | 24. (4) | 25. (4) |
| 26. (1) | 27. (2) | 28. (4) | 29. (2) | 30. (3) |
| 31. (3) | 32. (4) | 33. (2) | 34. (4) | 35. (4) |

Chapter 7

Exercises

Single Correct Answer Type

1. (2) Intensity is given by, $\frac{P}{A} = \frac{3.9 \times 10^{26}}{4\pi r^2} = 5.62 \times 10^7 \text{ W/m}^2$
2. (1) Electric or magnetic field has zero average value in e.m.f. waves
3. (3) $Z = \sqrt{\frac{\mu_r}{\epsilon_r}} \times \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{50}{2}} \times 376.6 \Omega = 1883 \Omega$
4. (1) $\nu = 1057 \text{ MHz}$
 $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{1057 \times 10^6} \text{ m} = 0.28 \text{ m} = 28 \text{ cm}$ (radio waves)
5. (2) $\lambda_m T = \text{constant}$
 $\Rightarrow \lambda_m \times (2.7 \text{ K}) = 0.2896 \text{ cm K}$
 $\therefore \lambda_m = \frac{0.2896}{2.7} \text{ cm} = 0.10 \text{ cm} = 1 \text{ mm}$
6. (3) $E = h\nu \therefore E \propto \nu$
7. (2) $E = \frac{hc}{\lambda}$, $E \propto \frac{1}{\lambda}$ or $eV \propto \frac{1}{\lambda}$
8. (1) X-rays are not reflected from the target, so cannot be used for radar.
9. (3) $E = h\nu$
 $V = E/h = 5 \times 10^{17} \text{ Hz}$
10. (4) $eV = h\nu = \frac{hc}{\lambda} \therefore \lambda = \frac{hc}{eV}$
11. (2) Area surrounding the TV tower is
 $A = \pi r^2$ ($r = d$)
 $= \pi d^2 = \pi (\sqrt{2hR})^2 = \pi \cdot 2hR$
 $a \propto h \therefore a' = 2a$, $h' = 2h$
12. (1) Here, $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{8.2 \times 10^6} = 36.6 \text{ m}$
13. (2) Comparing the given equation with equation of plane electromagnetic wave, $E_z = E_0 \cos(\omega t + kx)$, we have
 $\omega = 6 \times 10^8$ and $k = 4$.
 Velocity of light in medium
 $\nu = \frac{\omega}{k} = \frac{6 \times 10^8}{4} = \frac{3}{2} \times 10^8 \text{ m/s}$
 Refractive index, $\mu = \frac{c}{\nu} = \frac{3 \times 10^8}{(3/2) \times 10^8} = 2$
14. (3) $\mu = \frac{c}{\nu}$
 Now $c = \frac{1}{\sqrt{\mu_0 K_0}}$ and $\nu = \frac{1}{\sqrt{\mu_0 K_0 \mu_r K_r}}$
 $\therefore \mu = \sqrt{\mu_r K_r}$
 But $\mu_r = \mu_0$ and $K_r = K_0$
 $\therefore \mu = \sqrt{\mu_0 K_0}$

15. (2) $\mu = \frac{c}{\nu} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \times \frac{\sqrt{\mu \epsilon}}{1} = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}}$
16. (3) $c = \frac{E_0}{B_0}$
 $\therefore B_0 = \frac{E_0}{c} = 1.6 \times 10^{-6} \text{ Wb/m}^2$
17. (1) As $B \propto r$, since the point is on the axis, where $r = 0$, so $B = 0$.
18. (3) $I_{av} = \frac{1}{2} \epsilon_0 E_0^2 \times c = 1.719 \text{ W/m}^2$
19. (1) Average energy density of electromagnetic wave
 $U_{av} = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} \times (8.85 \times 10^{-12}) \times (50)^2$
 $\approx 10^{-8} \text{ J/m}^3$
20. (3) $C = \frac{\epsilon_0 K A}{d} = \frac{(8.85 \times 10^{-12}) \times 10 \times 1}{10^{-3}} = 8.85 \times 10^{-8} \text{ F}$
 $i = \frac{d}{dt}(CV) = C \frac{dV}{dt} = 8.85 \times 10^{-8} \times 25 = 2.2 \times 10^{-6} \text{ A}$
 $= 2.2 \mu\text{A}$
21. (2) Charge on capacitor plates at time t is, $q = it$
 Electric field between the plates at this instant.
 $E = \frac{q}{A \epsilon_0} = \frac{it}{A \epsilon_0}$
 Electric flux through the given area
 $\phi_E = \left(\frac{A}{2}\right) E = \frac{it}{2 \epsilon_0}$
 Therefore, displacement current
 $i_d = \epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 \frac{d}{dt} \left(\frac{it}{2 \epsilon_0}\right) = \frac{i}{2}$
22. (3) Total power = Solar constant $\times$ Area
 $= 10^4 \times (10 \times 10) \times 10^6 \text{ W}$
23. (4) Electric energy density
 $u_e = \frac{1}{2} \epsilon_0 E_{rms}^2$
 $E_{rms} = \frac{E_0}{\sqrt{2}}$
 $\therefore u_e = \frac{1}{4} \epsilon_0 E_0^2$
24. (4) Power = Area $\times$ Solar constant $= 1.62 \times 10^5 \text{ W}$
25. (4) Force of radiation $= \frac{\text{Power}}{c} = \frac{1.6 \times 10^5}{3 \times 10^8} = 5.32 \times 10^{-4} \text{ N}$
26. (1) Transferred momentum/second to mirror is
 $P = \frac{2AS}{C} = 1.21 \times 10^{-10} \text{ kg m s}^{-1}$
27. (2) Number of oscillation in coherence length
 $\frac{\ell}{\lambda} = \frac{0.024}{5900 \times 10^{-10}} = 4.068 \times 10^4$

$$28. (4) B = \frac{E}{c} = \frac{10^{-4}}{3 \times 10^8} = 3.3 \times 10^{-13} \text{ T}$$

$$29. (2) B_0 = \frac{E_0}{c} = \frac{10^{-3}}{3 \times 10^8} = 3.3 \times 10^{-12} \text{ T}$$

$$30. (3) \oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$$

$$\text{or } E 2\pi r = \frac{d}{dt}(Kt \times \pi r^2)$$

$$\text{or } E = \frac{Kr}{2}$$

31. (3) Radiation force = Momentum transferred per second by electromagnetic wave to the mirror

$$= \frac{2I_{\text{av}} A}{c} = \frac{2 \times (10) \times (20 \times 10^{-4})}{(3 \times 10^8)} = 1.33 \times 10^{-10} \text{ N}$$

$$32. (4) \text{ The pointing vector } \vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$= \frac{1}{4\pi \times 10^{-7}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 80 & 32 & -64 \\ 0.2 & 0.08 & 0.29 \end{vmatrix} = 11.52\hat{i} - 28.8\hat{j}$$

$$33. (2) \omega = 2\pi\nu \frac{2\pi c}{\lambda} = \frac{2\pi \times 3 \times 10^8}{6 \times 10^{-3}}$$

$$= \pi \times 10^{11} \text{ rad/s}$$

The equation for the electric field, along y-axis in the electromagnetic wave is

$$E_y = E_0 \sin \omega \left(t - \frac{x}{c} \right) = 33 \sin \pi \times 10^{11} (t - x/c)$$

$$34. (4) B_0 = \frac{E_0}{c} = \frac{48}{3 \times 10^8} = 16 \times 10^{-8} \text{ T}$$

$$B_{\text{rms}} = \frac{B_0}{\sqrt{2}} = \frac{16 \times 10^{-8}}{\sqrt{2}} = 8\sqrt{2} \times 10^{-8} \text{ T}$$

$$= 11.3 \times 10^{-8} \text{ T}$$

35. (4) Wave is propagating along positive X-axis; magnetic field is directed along Z-axis so electric field must be directed along $(\hat{k} \times \hat{i}) = \hat{j}$, along Y-axis.

Amplitude of electric field

$$E_0 = B_0 c = 2 \times 10^{-7} \times 3 \times 10^8 = 60 \text{ V/m}$$

So choice (4) is correct.

